UKAPA7 Layer U Chicago vs UK Comparison Report

This report compares the current UKAPA7 U-layer tension measurements from:

- UK factory JSON data in `UKAPA7U.json`
- Chicago summary CSV data in `data/tension_summaries/tension_summary_UKAPA7_U.csv`

Residual definition throughout: `Chicago - UK`.

Data Availability

- The UK JSON contains `1141` non-null tensions on both side A and side B.
- The current Chicago summary CSV contains `417` side-A values and `641` side-B values.
- All non-null Chicago values align to non-null UK values at the same wire number.
- The aligned wire ranges are `400` through `940` on side A and `241` through `967` on side B.

This is therefore a direct same-index comparison over the currently available Chicago subset on each side, not a full-layer comparison over all U wires.

Executive Summary

- Side A currently has `417` aligned wires. Its residuals are broadly negative, with mean `-0.600 N`, standard deviation `0.646 N`, and weak correlation (`-0.017`) between the Chicago and UK measurements.
- Side B currently has `641` aligned wires. It is also systematically negative, with mean residual `-0.871 N`, standard deviation `0.548 N`, and noticeably stronger correlation (`0.509`) than side A.
- Both sides therefore show Chicago tensions below the UK tensions on average, but side B is more coherent as a trace while side A is noisier and less correlated in the currently populated window.

Source Artifacts

- Comparison table: `data/tension_summaries/tension_comparison_UKAPA7_U.csv`
- Raw A-side plot: `data/tension_plots/tension_raw_A_UKAPA7_U.png`
- Raw B-side plot: `data/tension_plots/tension_raw_B_UKAPA7_U.png`

- Combined residual plot:
  `data/tension_plots/tension_residual_UKAPA7_U.png`

Summary Statistics

Side	Aligned Wires	Wire Range	Chicago Mean (N)	UK Mean (N)	Corr	Mean Diff (N)	Median Diff (N)	Std Diff (N)	MAE (N)	RMSE (N)	Negative Fraction
A	417	`400-940`	5.366	5.966	-0.017	-0.600	-0.515	0.646	0.685	0.881	0.837
B	641	`241-967`	5.331	6.202	0.509	-0.871	-0.810	0.548	0.895	1.029	0.972

Raw Side-A Comparison

![UKAPA7 U raw A](../data/tension_plots/tension_raw_A_UKAPA7_U.png){ width=100% }

Interpretation

- The Chicago A-side mean is about `0.600 N` below the UK A-side mean over the
  populated Chicago window.
- The trace-level correlation is currently very weak, so side A does not show a
  strong wire-by-wire match in this subset.
- The residuals are mostly negative, but they are wide enough that side A
  should be treated as a noisy comparison rather than a nearly constant shift.

Raw Side-B Comparison

![UKAPA7 U raw B](../data/tension_plots/tension_raw_B_UKAPA7_U.png){ width=100% }

Interpretation

- Side B shows a larger average Chicago-versus-UK deficit, about `0.871 N`.
- Unlike side A, side B preserves a visible trace shape, reflected in the
  stronger correlation (`0.509`).
- The negative fraction is `0.972`, so the B-side residuals are almost entirely
  below zero in the currently available Chicago window.

Residual Comparison

![UKAPA7 U residuals](../data/tension_plots/tension_residual_UKAPA7_U.png){ width=100% }

Interpretation

- Both sides are biased negative, consistent with Chicago measurements being

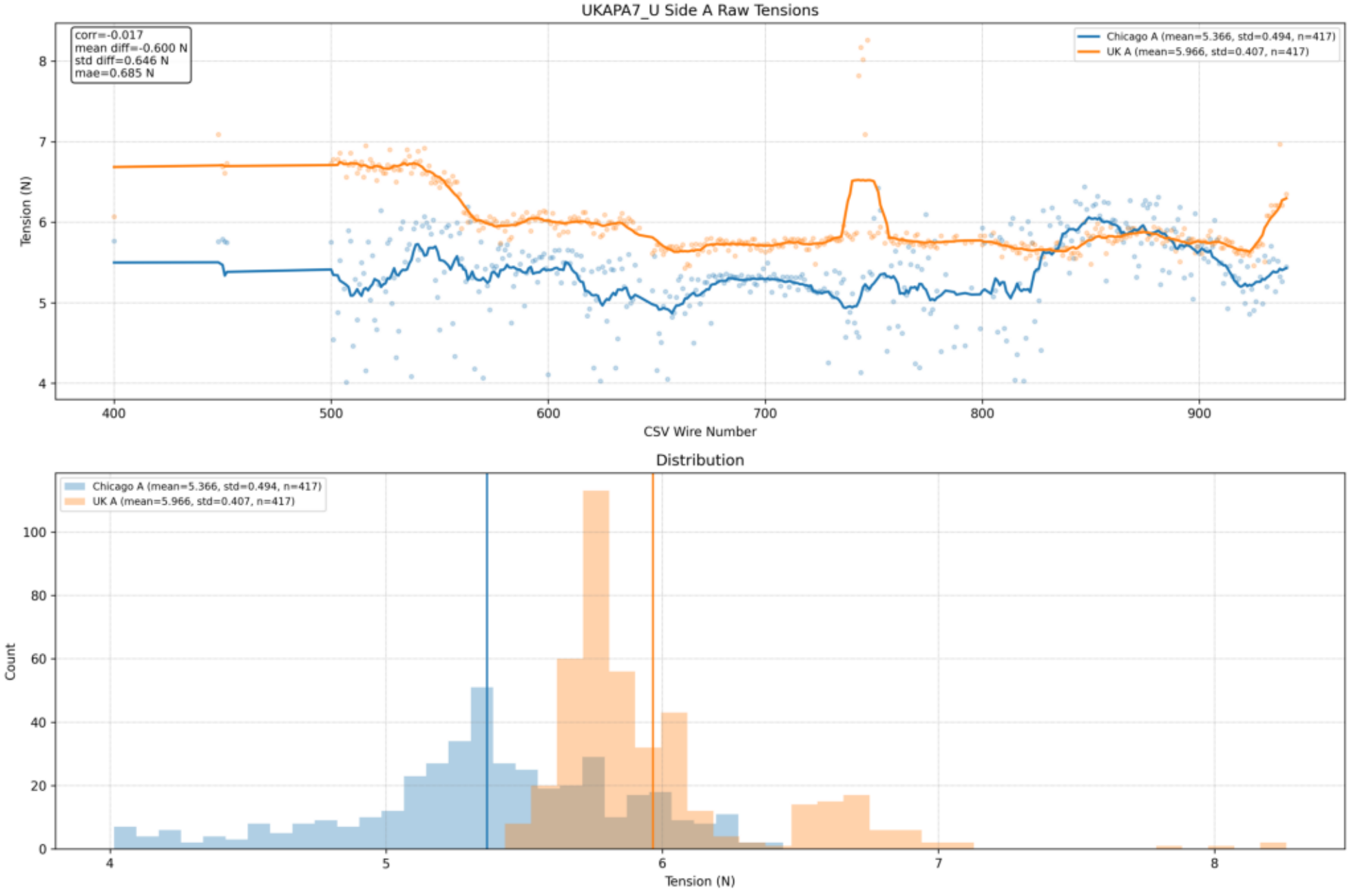
  lower than the corresponding UK measurements in the available U-layer data.
- Side A has the smaller mean shift but the broader and less structured residual pattern.
- Side B has the larger mean shift, but its residuals are more clustered and more consistent with the underlying UK trace.

Conclusion

The current UKAPA7 U-layer comparison shows the same sign on both sides: Chicago tensions are lower than UK tensions on average. Side A and side B do not behave equally, though. In the available A-side window, the comparison is noisy and only weakly correlated wire-by-wire. In the larger B-side window, the Chicago data track the UK structure more clearly, but with a larger overall negative offset.

This report is intentionally limited to the currently populated U-layer rows in the Chicago summary CSV. As more U-layer wires are added, the same plots and statistics should be regenerated before drawing stronger conclusions about the full layer.

tension_raw_A_UKAPA7_U.png



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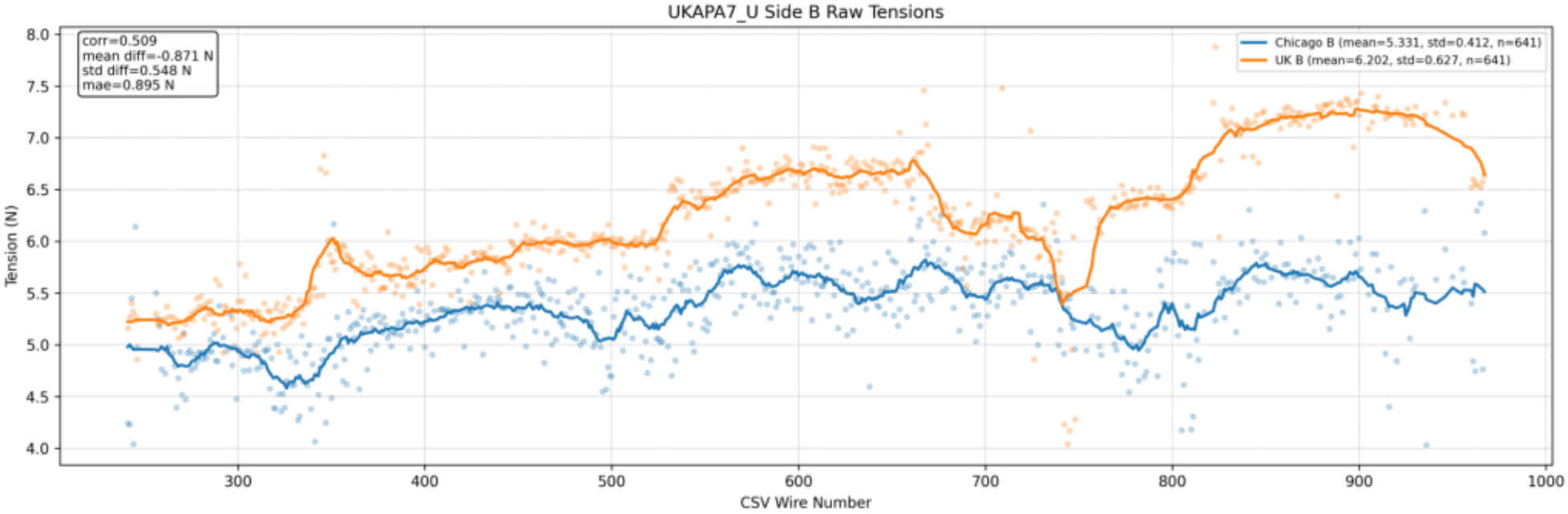

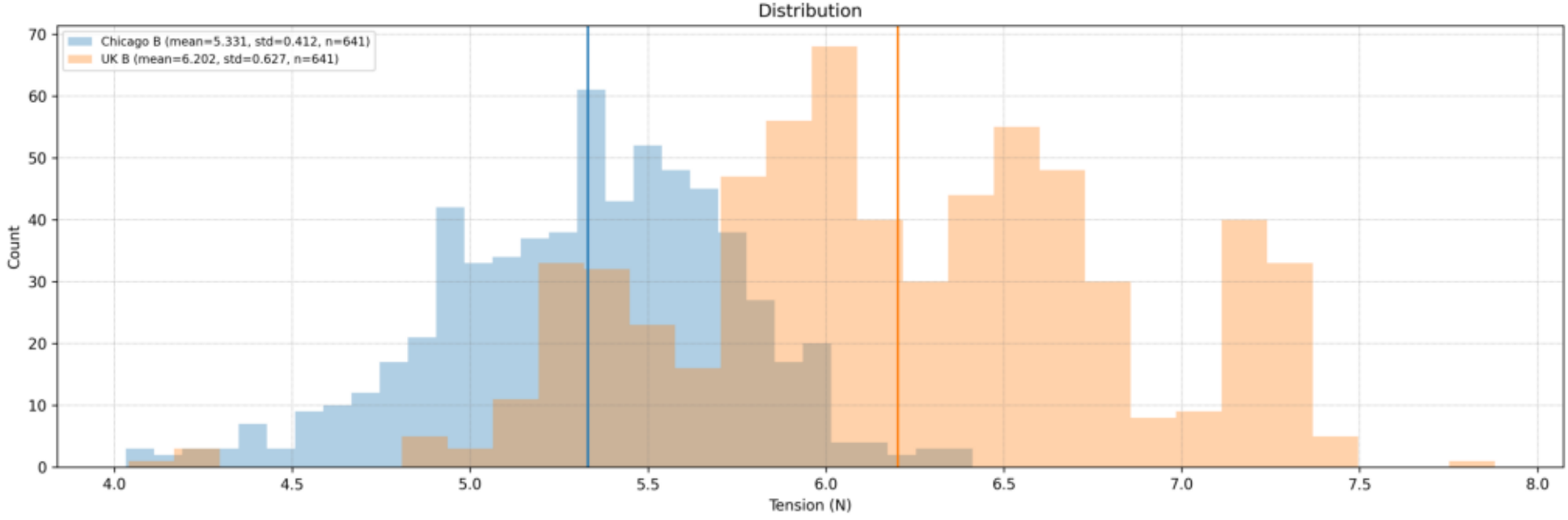
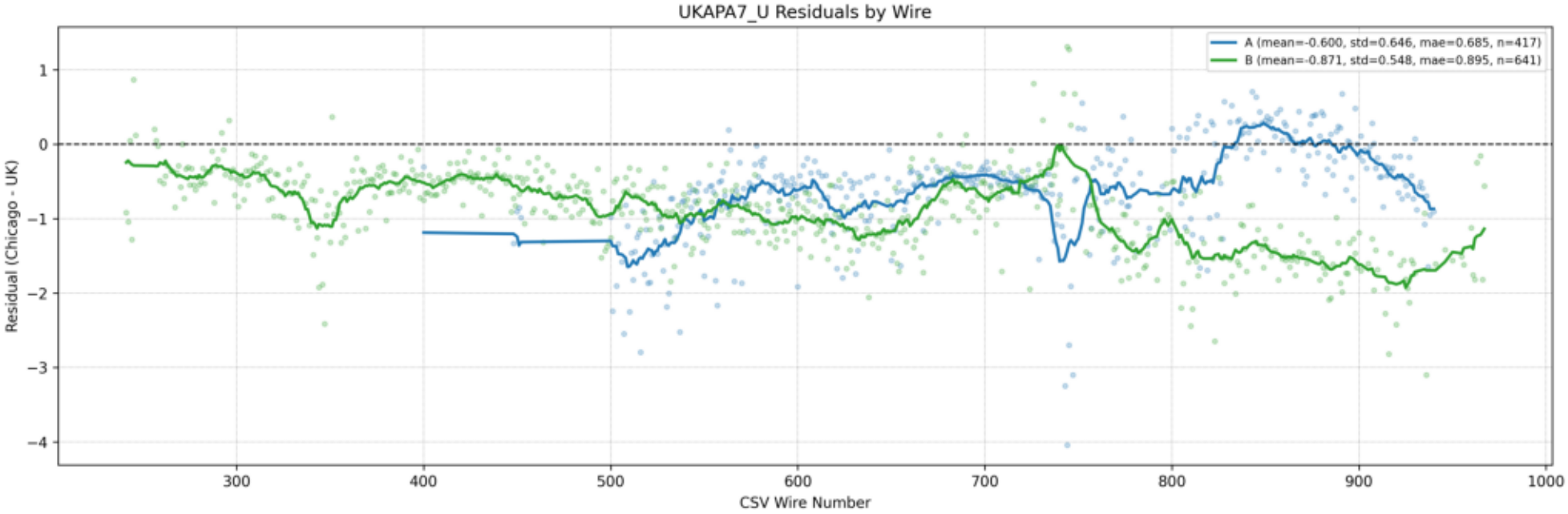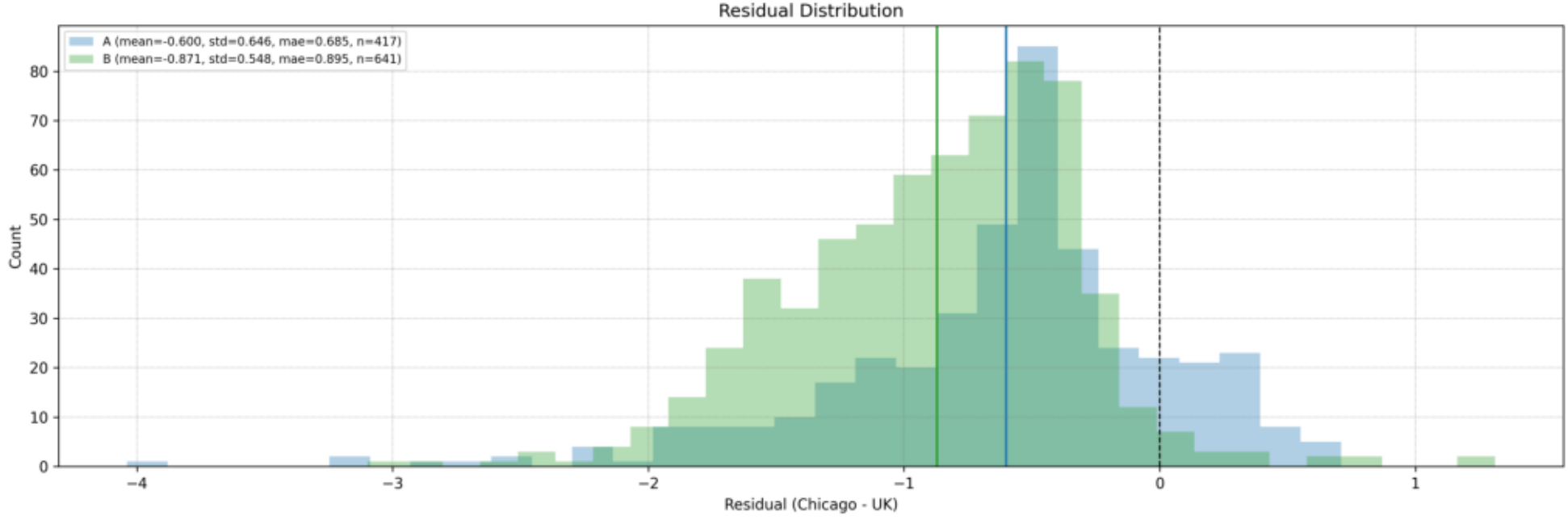
tension_residual_UKAPA7_U.png